HEALTHCARE ANALYTICS

MSBA 382

Diabetes Analytics Dashboard

A Consultant-to-Client Manual

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1-Summary:

The Diabetes Analytics Dashboard, a consulting-grade analytical tool designed to assist your institution in understanding diabetes severity, identifying at-risk patient profiles, and screening specific patients using a predictive risk model, is described in this manual. Using a live, active web application, the system connects clinical data at individual level with global epidemiological information.

The dashboard is divided into five parts: a worldwide prevalence map, a machine-learning-based risk predictor, a deep-dive clinical analysis, a demographic and risk-factor analysis, and an overview summary. This manual's Section 4 will give a detailed description of each.

2-Data sources:

The dashboard uses two complementing datasets to offer both more wide population-level background and detailed patient-level information.

2.1 The Pima Indian Diabetes Data:

The demographic data, clinical, and prediction elements of the dashboard are powered by this main health data.

- Source: Kaggle/UCI Machine Learning Repository, National Institute of Diabetes and Digestive and Kidney Diseases
- Population: 768 female patients
- Structure: 1 binary result (diabetic or non-diabetic), and 8 input variables
- Data quality: five variables had biologically unverified zero values, though there were no missing (null) values or duplicate data

2.2 IDF global diabetes Atlas:

The international and local background that supports the patient-level analysis is provided by this dataset

- Source: Diabetes Atlas, 10th Edition (2021), International Diabetes Federation (IDF)
- Coverage rate: diabetes rate (%) at the national level between individuals aging 20–79 in more than 80 countries

3-Table of variables:

Each clinical variable utilized in the dashboard is listed in the table below, which we used from the Pima Indian diabetes data.

Variable	**Description**	**Data Quality Note**
Pregnancies	The patient's total number of pregnancies	Clean
Glucose	2-hour oral glucose tolerance test and plasma glucose levels (mg/dL)	0.7% zero-values imputed
Blood Pressure	BP diastolic (mm Hg)	4.6% zero-values estimated
Skin Thickness	Skinfold thickness of the triceps (mm)	29.6% zero-values estimated
Insulin	Serum insulin levels after two hours (mu U/mL)	48.7% zero-values imputed
BMI	Body Mass Index: weight in kilograms divided by height in meters	1.4% zero-values estimated
Diabetes Pedigree Function	A function that uses family history to score the probability of diabetes	Clean
Age	Age of the person being treated in years	Clean
Outcome	Diagnosis of binary diabetes: 0 = non-diabetic, 1 = diabetic	Clean (target variable)

3.1 Methodology and Data Cleaning:

There were zero values in 5 clinical variables (BMI, insulin, skin thickness, blood pressure, and glucose) that are not medically feasible for a living human.

Instead of remaining empty these zeros indicate the missing information that was initially encoded as 0. Each zero-value was used with the variable's median value, which was determined independently for each column, in order to maintain the accuracy of data throughout all dashboard representations. Each of the dashboard sections, especially the data powering the ML model, experienced the same cleaning procedure.

4. dashboard components:

Available through the left-hand navigation sidebar, the dashboard is organized into five separate pages.

4.1 overview: The main Metrics are displayed on the landing page, including the number of patients overall, the prevalence of diabetes generally, and the mean clinical parameters (BMI, glucose) within patients with diabetes. A glucose distribution histogram distinguishing the two groups, donut chart visualization displaying the diabetic vs. non-diabetic difference, and international backdrop statistics (537 million people suffering from diabetes worldwide; 6.7 million annual mortality rates; a predicted 783 million cases by 2045) are also included.

4.2 demographics: This page examines the relationship between diabetes risk and age, pregnancy count, and BMI. A bar chart showing the diabetes rates by age category, a stacking bar chart showing patient statistics by age group, a line graph of the diabetes rate versus the amount of pregnant women, and a grouped bar chart contrasting the BMI groups (Underweight, Normal, Overweight, and Obese) among patients with and without diabetes are all included. The primary takeaway from the page is summed up in a crucial-insight callout box.

4.3clinical: Clinical factors can be explored interactively on this page. Box plots showing diabetic vs. non-diabetic normal distributions. and a scatter plot versus glucose can be created by the user selecting any clinical variable (Glucose, BMI, Blood Pressure, Insulin, Diabetes Pedigree Function, or Skin Density) using a dropdown option. A categorized bar chart of average clinical values by result, a correlation heatmap across all variables, and complete summary statistics data for each group of patients are also included on the page.

4.4global map: The patient-level results are placed in the context of global demography on this page. With a slider in order to filter nations above a particular rate threshold, this editable global chorus map shows the incidence of diabetes by nation. The 10 nations with the highest prevalence and the average prevalence by global region are listed in the following tables.

4.5Risk predictor: A random forest model of classification (100 trees, maximum depth of 8, 80/20 train-test split, normalized features) trained on the filtered clinical dataset is displayed on the last page. A real-time based diabetes risk estimate is shown as a percentile with a color-coded gauge chart, and clients can modify sliders for each of the eight medical inputs. The performance of the model is shown in a second tab along with feature priority rankings, a confusion matrix, and a complete classification report (precision, recall, F1-score). With held-out test data, the model's accuracy is 74%.

5-Going to the Dashboard

There is no need to install or log in because the dashboard is live.
• Diabetes-dashboard-fgolhtbyxfocgxzhtr2hwj.streamlit.app is the live URL.
• Hosting: a public GitHub repository that is linked to Streamlit Community Cloud
• Compatibility: available for both on desktop and mobile devices using any current web browser.

5.1 recommended use: we advise healthcare personnel to start with the Overview page to get to know themselves with overall incidence figures, then proceed to Demographic data and Clinical Analysis to determine which individual accounts require more attention to them.